

A K-SPACE DIRECTIVITY SOLVER FOR MULTI-CHANNEL PARAMETRIC ARRAY LOUDSPEAKERS

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ABSTRACT

Far-field directivity prediction is essential for parametric array loudspeakers (PALs). Conventional loudspeaker directivity can be derived from the angular spectrum of a prescribed aperture. In PALs, however, audible sound is generated by a volumetric virtual source, making aperture-based angular-spectrum analysis insufficient. We propose a k -space-shell solver that constructs this nonlinear source and samples its spatial Fourier spectrum on the audio-wavenumber shell, where each point corresponds to a far-field direction. By accounting for nonlinear interactions throughout the computational domain, the solver avoids the near-field coupling approximations used in the state-of-the-art modified convolution directivity model (MCDM). An observable-oriented dimension-reduction formulation and block-wise streaming strategy make large-scale simulations practical, requiring 2.00 GB of peak memory in a representative case. Comparisons show close agreement with MCDM for continuous apertures. For discrete multi-channel steered arrays, the proposed solver more accurately captures sidelobe and null structures by retaining nonlinear interactions approximated by MCDM.

Index Terms— Parametric array loudspeaker, far-field directivity, k -space method, nonlinear acoustics, multi-channel array

1. INTRODUCTION

Parametric array loudspeakers (PALs) generate audible sound through the nonlinear interaction of ultrasonic primary waves [1–7]. Owing to their narrow audio beams and compact ultrasonic apertures, PALs have been studied for directional audio reproduction, personal sound zones, and spatial audio applications [3, 8, 9]. Recent multi-channel PAL systems further introduce channel-wise amplitude and phase control, enabling electronic steering and more flexible transmit-side sound-field design [9–14]. This additional flexibility, however, also makes audio far-field directivity prediction more difficult because the generated field is no longer determined by a simple linear array factor [15, 16].

Unlike conventional loudspeaker arrays, PALs rely on nonlinear self-demodulation, and the audible field is produced by a distributed virtual source whose spatial structure depends on ultrasonic propagation, aperture geometry, element discreteness, and multi-channel excitation [1–3, 17]. For practical discrete arrays, steering phases and ultrasonic grating lobes can produce complex virtual-source structures that directly affect the resulting audio directivity [15–17]. A useful simulator should therefore connect the ultrasonic transmit configuration to the far-field audio field through the intermediate virtual-source distribution.

Several analytical and semi-analytical models have been developed for PAL directivity prediction [18, 19]. Convolution-based models relate the audio directivity to the directivities of the ultrasonic primary fields and provide efficient approximations [15, 16]. However, for discrete multi-channel arrays, steering-dependent interference and element layout can produce spatial structures that are difficult to represent using simplified models. A direct numerical formulation that preserves the virtual-source distribution is therefore desirable.

Existing k -space methods model ultrasonic propagation and nonlinear sound generation in finite near-field domains [17, 20], but do not directly provide far-field directivity. Extending such simulations to far-field distances requires prohibitively large three-dimensional domains. We therefore derive far-field directivity from the spatial Fourier spectrum of the nonlinear virtual source on the audio-wavenumber shell. For prescribed observation planes, observable-oriented reduction evaluates only the required directions, while block-wise accumulation avoids storing the complete virtual source.

2. THEORY

2.1. k -space directivity formulation

Let the two ultrasonic primary pressures be $p_1(\mathbf{r}, \omega_1)$ and $p_2(\mathbf{r}, \omega_2)$. The generated audio difference frequency is $\omega_a = \omega_2 - \omega_1$, with $k_a = \omega_a/c_0$. Under the quasi-linear approximation [1, 3], the equivalent audio virtual source is

$$q_a(\mathbf{r}, \omega_a) = \frac{\beta \omega_a}{i \rho_0^2 c_0^4} p_1^*(\mathbf{r}, \omega_1) p_2(\mathbf{r}, \omega_2). \quad (1)$$

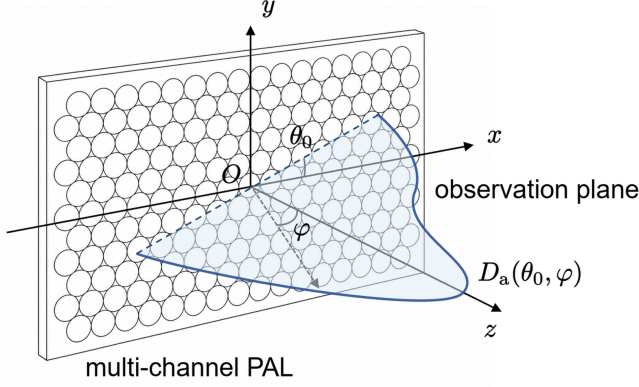


Fig. 1. Geometry of the multi-channel PAL aperture and the prescribed observation plane for evaluating $D_a(\theta_0, \varphi)$.

The audio pressure generated by the virtual source can be expressed using the free-space Green's function as

$$p_a(\mathbf{r}, \omega_a) = -i\rho_0\omega_a \iiint_{\mathbb{R}^3} q_a(\mathbf{r}_v, \omega_a) \frac{e^{ik_a|\mathbf{r}-\mathbf{r}_v|}}{4\pi|\mathbf{r}-\mathbf{r}_v|} d^3\mathbf{r}_v, \quad (2)$$

where $\mathbb{R}^3$ denotes the three-dimensional free-space domain.

For far-field observation, $|\mathbf{r} - \mathbf{r}_v| \approx r - \hat{\mathbf{s}} \cdot \mathbf{r}_v$, where $\hat{\mathbf{s}} = \mathbf{r}/r$. The pressure then becomes

$$p_a(\mathbf{r}, \omega_a) = \frac{e^{ik_a r}}{r} D_a(\hat{\mathbf{s}}, \omega_a), \quad (3)$$

where the audio directivity is

$$D_a(\hat{\mathbf{s}}, \omega_a) = -\frac{i\rho_0\omega_a}{4\pi} \iiint_{\mathbb{R}^3} q_a(\mathbf{r}_v, \omega_a) e^{-ik_a\hat{\mathbf{s}} \cdot \mathbf{r}_v} d^3\mathbf{r}_v. \quad (4)$$

Equation (4) shows that the far-field directivity is determined by the spatial spectrum of the virtual source evaluated on the audio-wavenumber shell,

$$D_a(\hat{\mathbf{s}}, \omega_a) = -\frac{i\rho_0\omega_a}{4\pi} \tilde{q}_a(\mathbf{k}, \omega_a) \Big|_{\mathbf{k}=k_a\hat{\mathbf{s}}}, \quad (5)$$

where the spatial Fourier transform of the virtual source is

$$\tilde{q}_a(\mathbf{k}, \omega_a) = \iiint_{\mathbb{R}^3} q_a(\mathbf{r}_v, \omega_a) e^{-i\mathbf{k} \cdot \mathbf{r}_v} d^3\mathbf{r}_v. \quad (6)$$

This formulation connects the PAL transmit configuration with the far-field audio directivity through the intermediate virtual-source distribution. In the proposed solver, the primary ultrasonic fields are calculated first, the virtual source is then constructed according to Eq. (1), and the far-field directivity is obtained by evaluating the spatial spectrum of the virtual source on the audio-wavenumber shell.

2.2. Observable-oriented reduction and streaming

In practical PAL applications, directivity is typically examined on one or a few observation planes, such as the xOz and yOz planes, because beam steering and coverage are usually designed within these planes. Computing the directivity over the full sphere is therefore unnecessary in some practical scenarios. As shown in Fig. 1, the proposed solver evaluates only the directions in a prescribed observation plane perpendicular to the xOy plane, whose intersection with the xOy plane is specified by θ_0 . This observable-oriented setting allows the virtual-source distribution to be reduced before the final spatial-spectrum evaluation. The observation direction is parameterized as

$$\hat{\mathbf{s}}(\theta_0, \varphi) = (\sin \varphi \cos \theta_0, \sin \varphi \sin \theta_0, \cos \varphi), \quad (7)$$

where φ is the scanning angle within the prescribed observation plane.

Taking the xOz plane as a representative case, namely $\theta_0 = 0$, the observation direction becomes $\hat{\mathbf{s}}(0, \varphi) = (\sin \varphi, 0, \cos \varphi)$. Since the observation direction has no y component, the virtual source can be projected along y as

$$q_{xz}(x, z, \omega_a) = \int_{-\infty}^{+\infty} q_a(x, y, z, \omega_a) dy. \quad (8)$$

The far-field directivity in this observation plane can then be evaluated from the reduced source distribution rather than from the complete three-dimensional virtual source. The same principle can be applied to other prescribed observation planes by selecting the corresponding projection direction.

The computational domain required for virtual-source convergence can be very large. Therefore, as illustrated in Fig. 2, the proposed implementation processes the domain in axial blocks along the propagation direction. For the i -th block, the primary fields are calculated, the local virtual source is generated and projected onto the prescribed observation plane, and its contribution to the far-field directivity is accumulated,

$$p_1^{(i)}, p_2^{(i)} \rightarrow q_a^{(i)} \rightarrow q_{xz}^{(i)} \rightarrow D_a^{(i)}. \quad (9)$$

Owing to the linearity of the Fourier projection, the final directivity is obtained by summing the contributions from all axial blocks,

$$D_a(\theta_0, \varphi, \omega_a) = \sum_{i=1}^{N_b} D_a^{(i)}(\theta_0, \varphi, \omega_a). \quad (10)$$

Only the current three-dimensional block, its temporary two-dimensional projection, and the accumulated far-field directivity are stored during this process. The observable-oriented reduction and block-wise streaming strategy therefore avoid storing the complete three-dimensional virtual source while preserving its contribution to the prescribed far-field observation plane.

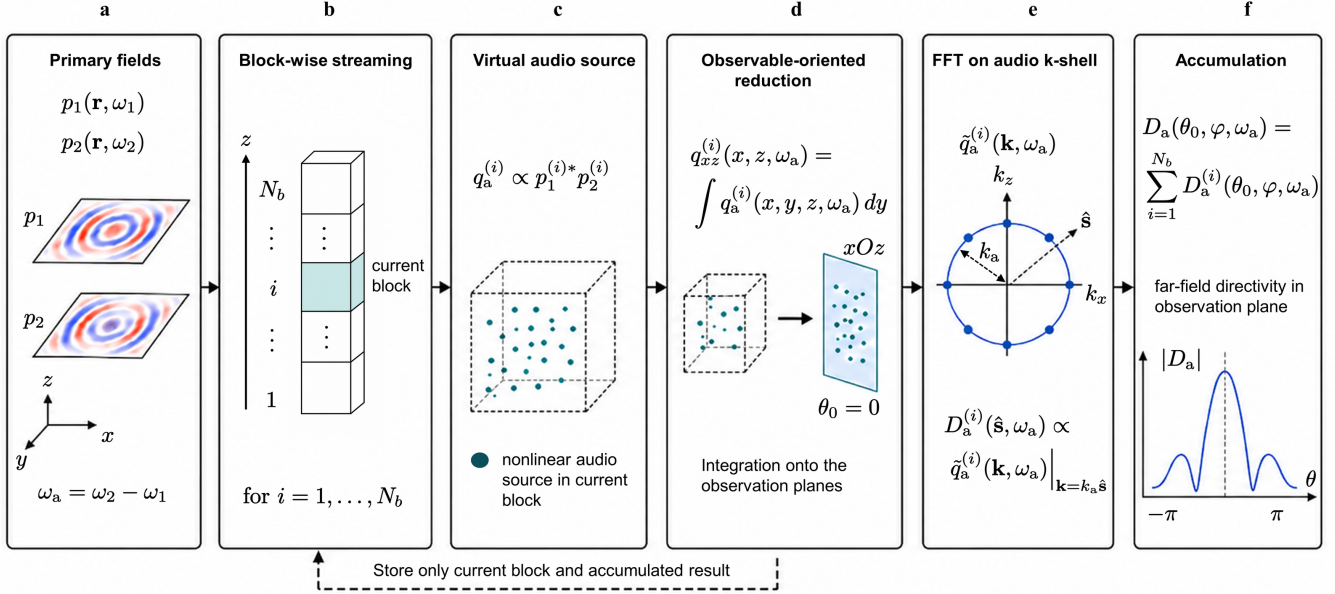


Fig. 2. Workflow of the proposed solver: block-wise ultrasonic propagation, virtual-source generation, observation-plane reduction, and k -space far-field evaluation.

3. NUMERICAL VALIDATION AND COMPUTATIONAL PERFORMANCE

3.1. Simulation-domain convergence

The computational domain is determined using a rectangular PAL aperture with $W = 0.165$ m and $H = 0.109$ m. The primary frequencies are $f_1 = 40$ kHz and $f_2 = 41$ kHz, giving $f_a = 1$ kHz. The domain is $x \in [-L_x/2, L_x/2]$, $y \in [-L_y/2, L_y/2]$, and $z \in [0, L_z]$, with $\Delta x = \Delta y = 0.01$ m and $\Delta z = 0.005$ m. In Fig. 3, $z_{\max} = L_z$ and $L_x = L_y = L_{xy}$.

Convergence is assessed relative to the largest-domain result at $\varphi = 20^\circ, 40^\circ$, and 60° . As shown in Fig. 3(a) and (b), the level differences fall below 0.3 dB for $L_z \geq 24$ m. With $L_z = 24$ m, panels (c) and (d) show transverse convergence at approximately $L_{xy} = 1.4$ m. For steered sources, the domain is further expanded along the steering direction to cover the tilted virtual source.

3.2. Computational efficiency improvement

The proposed solver combines observable-oriented dimension reduction with block-wise streaming. A direct three-dimensional implementation stores the complete virtual source, whereas the proposed implementation projects each source block onto the observation plane and immediately accumulates its far-field contribution.

For the representative xOz observation plane, direct virtual-source construction and Fourier-shell evaluation require $O(N_x N_y N_z) + O(N_x N_y N_z N_\varphi)$ operations. Dimen-

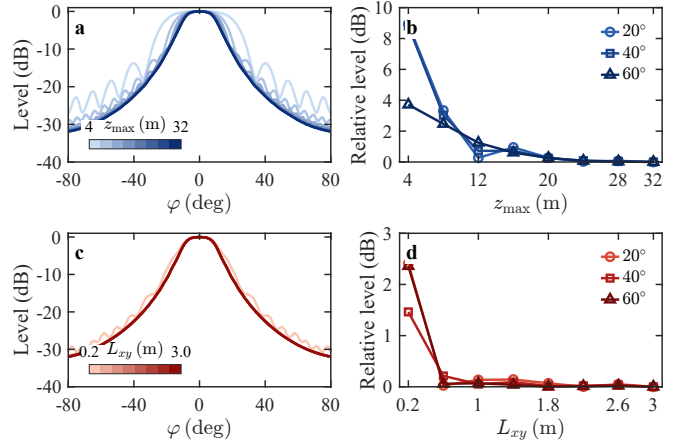


Fig. 3. Domain convergence for the rectangular PAL aperture. (a) Directivity versus $z_{\max}$ at $L_{xy} = 2$ m. (b) Selected-angle errors versus $z_{\max}$. (c) Directivity versus L_{xy} at $L_z = 24$ m. (d) Selected-angle errors versus L_{xy} .

sion reduction changes this to $O(N_x N_y N_z) + O(N_x N_z N_\varphi)$, reducing the Fourier-shell evaluation by a factor of N_y . Streaming further reduces the dominant memory requirement from $O(N_x N_y N_z)$ to $O(N_x N_y N_{z,b})$, or approximately by $N_z/N_{z,b}$, where $N_{z,b} = 64$. These estimates exclude ultrasonic propagation, which is common to both implementations.

For $L_z = 24$ m and $L_{xy} = 2$ m, the proposed implementation achieves a $3.21 \times$ speedup and reduces peak memory by

Table 1. Measured cost for two representative domains.

Implementation	L_z (m)	L_{xy} (m)	Runtime (s)	Peak memory (GB)
Direct 3-D	24	2	136.92	5.97
Observable-oriented streaming	24	2	42.69	2.00
Direct 3-D	24	4	—	$> 45.8^\dagger$
Observable-oriented streaming	24	4	167.13	2.44

† The calculation failed when requesting an internal array larger than 45.8 GB.

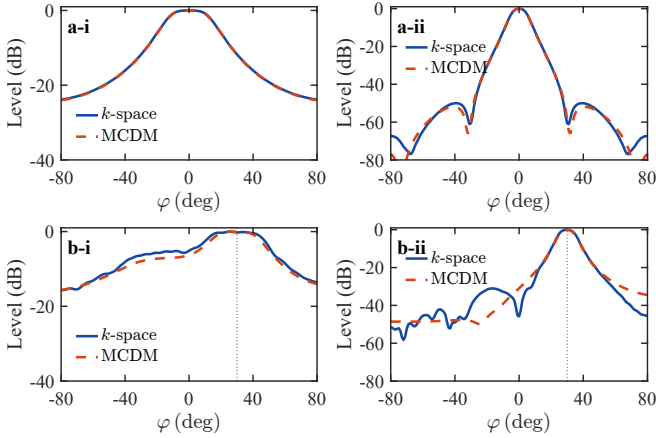


Fig. 4. Frequency-dependent comparison with MCDM in the main observation plane. (a-i, a-ii) Continuous aperture at $f_a = 500$ Hz and 4 kHz. (b-i, b-ii) Steered array at the same frequencies.

66.5%. At $L_{xy} = 4$ m, it remains executable with 2.44 GB, whereas the direct implementation cannot be completed on the same workstation. The proposed solver therefore supports large-domain PAL simulations on a standard desktop PC.

3.3. Comparison with MCDM

The proposed k -space-shell solver is compared with the modified convolution directivity model (MCDM) [16]. Because it accounts for the full spatial distribution of the virtual source within the converged domain without approximating near-field nonlinear coupling, the k -space result is used as the numerical reference.

Two sources are considered: a non-steered circular aperture with $a = 0.10$ m and a 32-channel staggered array steered to $\theta_s = 30^\circ$. Each channel drives one column of six circular elements with $a = 5$ mm. For both sources, $f_1 = 40$ kHz and $f_2 = f_1 + f_a$. The continuous aperture uses $(L_x, L_y, L_z) = (2.4, 2.4, 30)$ m, whereas the steered array uses $(L_x, L_y, L_z) = (36, 0.96, 36)$ m.

Figure 4 considers the main observation plane, $\theta_0 = 0^\circ$.

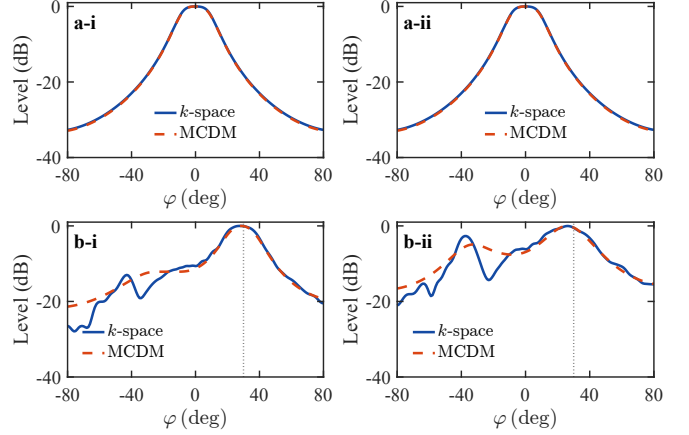


Fig. 5. Observation-plane comparison at $f_a = 1$ kHz. (a-i, a-ii) Continuous aperture at $\theta_0 = 15^\circ$ and 30° . (b-i, b-ii) Steered array in the same observation planes.

For the continuous aperture, the maximum errors above -20 dB remain below 0.5 dB. For the discrete array, they are 1.8 dB and 0.7 dB at 500 Hz and 4 kHz, respectively. At 4 kHz, the error above -30 dB reaches 5.1 dB because MCDM does not reproduce several sidelobe and directivity-null features.

For the continuous aperture in Fig. 5, the errors above -20 dB remain below 0.2 dB in both planes. For the discrete array, they increase to 6.4 dB and 8.6 dB at $\theta_0 = 15^\circ$ and 30° , respectively. By retaining spatial nonlinear interactions simplified in MCDM, the proposed solver more accurately captures observation-plane-dependent sidelobe and null structures.

4. CONCLUSION

This paper developed a spatial Fourier-domain k -space-shell solver for far-field directivity prediction of parametric array loudspeakers. By constructing the distributed nonlinear virtual source from the ultrasonic primary fields and evaluating its spatial spectrum on the audio-wavenumber shell, the proposed method directly links multi-channel transducer excitation to far-field audio radiation while retaining nonlinear interactions throughout the computational domain. An observable-oriented dimension-reduction formulation and block-wise streaming strategy were further introduced to avoid storing the complete three-dimensional virtual source, enabling large-domain simulations on a standard desktop PC. Numerical results verified convergence with respect to the axial and transverse computational domains and showed close agreement with MCDM for continuous apertures. For discrete multi-channel steered arrays, the proposed solver more accurately captured observation-plane-dependent sidelobe patterns and the locations and depths of directivity nulls by retaining spatial nonlinear interactions simplified in MCDM.

5. COMPLIANCE WITH ETHICAL STANDARDS

No ethical approval was required for this study. The authors have no relevant financial or non-financial interests to disclose.

6. ACKNOWLEDGMENT

This work was supported by the National Natural Science Foundation of China under Grants 12604712.

7. REFERENCES

- [1] P. J. Westervelt, "Parametric acoustic array," *J. Acoust. Soc. Am.*, vol. 35, no. 4, pp. 535–537, 1963.
- [2] M. B. Bennett and D. T. Blackstock, "Parametric array in air," *J. Acoust. Soc. Am.*, vol. 57, no. 3, pp. 562–568, 1975.
- [3] W.-S. Gan, J. Yang, and T. Kamakura, "A review of parametric acoustic array in air," *Appl. Acoust.*, vol. 73, no. 12, pp. 1211–1219, 2012.
- [4] P. Ji, E.-L. Tan, W.-S. Gan, and J. Yang, "A comparative analysis of preprocessing methods for the parametric loudspeaker based on the Khokhlov–Zabolotskaya–Kuznetsov equation for speech reproduction," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 19, no. 4, pp. 937–946, 2011.
- [5] J. Zhong, R. Kirby, and X. Qiu, "The near field, Westervelt far field, and inverse-law far field of the audio sound generated by parametric array loudspeakers," *J. Acoust. Soc. Am.*, vol. 149, no. 3, pp. 1524–1535, 2021.
- [6] T. Zhuang, J.-X. Zhong, and J. Lu, "The feasibility of sound zone control using an array of parametric array loudspeakers," in *Proc. Int. Symp. Chinese Spoken Lang. Process. (ISCSLP)*, 2024, pp. 66–70.
- [7] T. Zhuang, S. Li, F. Niu, J.-X. Zhong, and J. Lu, "Generating localized audible zones using a single-channel parametric loudspeaker," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2026, pp. 15 302–15 306.
- [8] C. Shi, Y. Kajikawa, and W.-S. Gan, "An overview of directivity control methods of the parametric array loudspeaker," *APSIPA Trans. Signal Inf. Process.*, vol. 3, p. e20, 2014.
- [9] K. Nakagawa, C. Shi, and Y. Kajikawa, "Beam steering of portable parametric array loudspeaker," in *Proc. Asia-Pacific Signal Inf. Process. Assoc. Annu. Summit Conf. (APSIPA ASC)*, Lanzhou, China, 2019, pp. 1824–1827.
- [10] W.-S. Gan, J. Yang, K.-S. Tan, and M.-H. Er, "A digital beamsteerer for difference frequency in a parametric array," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 14, no. 3, pp. 1018–1025, 2006.
- [11] C. Shi, "Investigation of the steerable parametric loudspeaker based on phased array techniques," Ph.D. dissertation, Nanyang Technol. Univ., Singapore, 2013.
- [12] Y. Zhu, Y. Zhang, F. Fan, W. Ma, L. Qin, Z. Kuang, M. Wu, and J. Yang, "An enhanced beamsteering algorithm based on MVDR for a multi-channel parametric array loudspeaker array," *J. Sound Vib.*, vol. 595, p. 118768, 2025.
- [13] Y. Zhu, L. Qin, W. Ma, F. Fan, M. Wu, Z. Kuang, and J. Yang, "A nonlinear sound field control method for a multi-channel parametric array loudspeaker array," *J. Acoust. Soc. Am.*, vol. 157, no. 2, pp. 962–975, 2025.
- [14] X. Wu, T. Zhuang, S. Li, M. Li, J.-X. Zhong, and J. Lu, "A comparative analysis of time-delay strategies for phased-array parametric loudspeakers," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 34, pp. 3743–3753, 2026.
- [15] C. Shi and Y. Kajikawa, "A convolution model for computing the far-field directivity of a parametric loudspeaker array," *J. Acoust. Soc. Am.*, vol. 137, no. 2, pp. 777–784, 2015.
- [16] J. Zhong, H. Zou, J. Lu, and D. Zhang, "A modified convolution model for calculating the far field directivity of a parametric array loudspeaker," *J. Acoust. Soc. Am.*, vol. 153, no. 3, pp. 1439–1451, 2023.
- [17] T. Zhuang, L. He, F. Niu, J.-X. Zhong, and J. Lu, "A k -space approach to modeling multi-channel parametric array loudspeaker systems," *J. Acoust. Soc. Am.*, vol. 158, no. 6, pp. 4651–4661, 2025.
- [18] S. Mei, Y. Hu, M. Alkahtani, and M. Nasr Esfahani, "A dual convolution model for grating lobe prediction in directivity of parametric array loudspeakers," *IEEE Access*, vol. 13, pp. 39 183–39 191, 2025.
- [19] C. Shi, Y. Wang, H. Xiao, and H. Li, "Extended convolution model for computing the far-field directivity of an amplitude-modulated parametric loudspeaker," *J. Phys. D: Appl. Phys.*, vol. 55, no. 24, p. 244002, 2022.
- [20] Y. Jing, M. Tao, and G. T. Clement, "Evaluation of a wave-vector-frequency-domain method for nonlinear wave propagation," *J. Acoust. Soc. Am.*, vol. 129, no. 1, pp. 32–46, 2011.